New lower bounds for hypergraph Ramsey numbers from affine-invariant colourings

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Abstract

We prove

$$R(4,5;3) \geq 38, \quad R(5,5;3) \geq 98, \quad R(4,6;3) \geq 68, \quad R(5,5;4) \geq 38, \quad R(5,6;4) \geq 98,$$

improving the lower bounds 35, 88, 63, 35 and 67 listed in revision #18 (April 2026) of Radziszowski's dynamic survey *Small Ramsey Numbers*. Each bound is witnessed by an explicit red/blue colouring of the r-subsets of a prime field $\mathrm{GF}(q)$ that is invariant under a group of affine maps $x \mapsto ax + b$, where a ranges over a fixed subgroup of $\mathbb{F}_q^*$ (the cube roots of unity for $R(4,5;3)$, the nonzero squares for $R(5,5;3)$ and $R(4,6;3)$, all of $\mathbb{F}_q^*$ for the 4-uniform cases). Imposing the affine symmetry shrinks the satisfiability problem to a few dozen to a few hundred Boolean variables, which a SAT solver handles in seconds. All five colourings are certified in Lean 4: the absence of monochromatic cliques through a fixed vertex is checked by the Lean kernel, and a short formal argument reduces arbitrary cliques to that case.

1 Introduction

For integers $k, l \geq r \geq 2$ the hypergraph Ramsey number $R(k,l;r)$ is the least n such that every colouring of the r-element subsets of an n-element set with the colours red and blue contains k points all of whose r-subsets are red, or l points all of whose r-subsets are blue. Such sets of points are called a red $K_k^{(r)}$ and a blue $K_l^{(r)}$. For $r \geq 3$ essentially no values are known: the only nontrivial classical value is $R(4,4;3) = 13$ [7], and Section 7.1 of the survey [1] lists only lower bounds for the next cases,

$$35 \leq R(4,5;3), \quad 63 \leq R(4,6;3), \quad 88 \leq R(5,5;3), \quad 79 \leq R(4,4,4;3), \quad 35 \leq R(5,5;4),$$

together with bounds such as $R(5,6;4) \geq 67$ obtained from general recurrences ([1, 7.1(f),(j)], [6]). The bounds for $R(4,5;3)$, $R(4,6;3)$, $R(5,5;3)$ and $R(4,4,4;3)$ are due to Dybizbański [2, 3], who found *cyclic* (shift-invariant) colourings with a SAT solver; before that $R(5,5;3) \geq 82$ was known [4]. The bound $R(5,5;4) \geq 35$ is due to Exoo [1].

In 2026 an unrefereed public project [8] used group-invariant colourings and SAT to obtain $R(4,4,4;3) \geq 84$, $R(4,6;3) \geq 64$ and $R(5,5;4) \geq 36$. For $R(5,5;3)$ that project lists experiments only at $n = 88$, and an issue in its tracker (September 2026, self-reported and not reviewed) claims an 88-point colouring, i.e. $R(5,5;3) \geq 89$.

Theorem 1.1. *$R(4,5;3) \geq 38$, $R(5,5;3) \geq 98$, $R(4,6;3) \geq 68$, $R(5,5;4) \geq 38$ and $R(5,6;4) \geq 98$.*

To our knowledge (literature and web search of October 14, 2026) all five bounds are new, and each improves every bound we are aware of by at least two. The case $R(4,5;3)$ is the smallest open classical hypergraph Ramsey number; its first lower bound, 24, is due to Isbell, and the bound 33 was improved to 35 by Dybizbański in 2018 [2, 1]. The proof is a construction: for a prime q we colour the r-subsets of $\mathrm{GF}(q)$ invariantly under a group of affine maps. Table 1 summarises the witnesses.

number	DS1.18	other 2026 claims	this paper	q	symmetry group
$R(4,5;3)$	≥ 35	—	≥ 38	37	$x \mapsto ax+b$, $a^3=1$
$R(5,5;3)$	≥ 88	≥ 89 (unreviewed)	≥ 98	97	$x \mapsto ax+b$, $a \in (\mathbb{F}_{97}^*)^2$
$R(4,6;3)$	≥ 63	≥ 64	≥ 68	67	$x \mapsto ax+b$, $a \in (\mathbb{F}_{67}^*)^2$
$R(5,5;4)$	≥ 35	≥ 36	≥ 38	37	$\mathrm{AGL}(1,37)$
$R(5,6;4)$	≥ 67	—	≥ 98	97	$\mathrm{AGL}(1,97)$

Table 1: The improved bounds. "Other 2026 claims" refers to [8].

2 Affine-invariant colourings

Let q be a prime and let $C \leq \mathbb{F}_q^*$ be a subgroup of order m. The group $G = G(q,C) = \{x \mapsto ax+b : \ a \in C,\ b \in \mathbb{F}_q\}$ acts on $\mathbb{F}_q$, transitively (already the translations do), and hence on the r-subsets of $\mathbb{F}_q$. A colouring χ of the r-subsets is *G-invariant* if $\chi(gS) = \chi(S)$ for all $g \in G$; such colourings correspond to colourings of the G-orbits of r-subsets, of which there are roughly $\binom{q}{r}/(qm)$.

Lemma 2.1. *Let χ be invariant under all translations $x \mapsto x+b$ of $\mathbb{F}_q$. If no k-subset containing 0 is monochromatic, then no k-subset is monochromatic.*

Proof. If S is monochromatic and $s \in S$, then $S-s$ contains 0 and is monochromatic of the same colour, because $\chi(T-s) = \chi(T)$ for every r-subset $T \subseteq S$. □

Consequently the existence of a G-invariant colouring without red $K_k^{(r)}$ and without blue $K_l^{(r)}$ is equivalent to the satisfiability of the CNF formula with one Boolean variable x_O per orbit O (true = red) and the clauses

$$\bigvee_{T \in \binom{S}{r}} \neg x_{O(T)} \quad (S \ni 0,\ |S| = k), \qquad \bigvee_{T \in \binom{S}{r}} x_{O(T)} \quad (S \ni 0,\ |S| = l).$$

Because the stabiliser of 0 in G (multiplication by C) permutes these clauses, many of them coincide; after removing duplicates, the formulas for $m = (q-1)/2$ have only 30–40 variables and at most a few times 10^4 clauses (for instance 34 variables for $q = 97$), and are decided by the SAT solver `kissat` [9] within seconds. We swept primes (and a few prime powers) q above the known bounds and divisors m of $q-1$ from large to small; Section 5 lists what was tried.

3 The constructions

Fix a prime q, the subgroup $C = (\mathbb{F}_q^*)^2$ of nonzero squares and a non-square g. If χ is a $G(q, C)$-invariant colouring of the triples of $\mathbb{F}_q$, then for any labelling $\{x, y, z\}$ of a triple,

$$\chi\{x,y,z\} = \begin{cases} \chi\{0,1,t\}, & t = \frac{z-x}{y-x}, \text{ if } y-x \text{ is a square,} \\ \chi\{0,g,t\}, & t = g\,\frac{z-x}{y-x}, \text{ if } y-x \text{ is a non-square.} \end{cases}$$

Hence χ is determined by the two sets $R_1 = \{t : \chi\{0,1,t\} = \text{red}\}$ and $R_g = \{t : \chi\{0,g,t\} = \text{red}\}$.

$R(4,5;3) \geq 38$. $q = 37$ and $C = \{1, 10, 26\}$, the cube roots of unity in $\mathbb{F}_{37}$ ($|G| = 111$, 78 orbits on triples), so the colouring is invariant under $x \mapsto x+1$ and $x \mapsto 10x$. Since C is small we describe the colouring by the colours of the triples $\{0, y, z\}$, $0 < y < z < 37$, which determine everything by translation: $\{x, y, z\}$ ($x < y < z$) has the colour of $\{0, y-x, z-x\}$. Exactly 258 of these 630 triples are red; they are listed in `data/describe_R453_37.txt` and encoded in the Lean file. In total 3182 of the 7770 triples are red; there is no red $K_4^{(3)}$ and no blue $K_5^{(3)}$ (this was also checked by brute force over all $\binom{37}{4}$ four-sets and $\binom{37}{5}$ five-sets).

$R(5,5;3) \geq 98$. $q = 97$, $g = 5$,

$$\begin{aligned} R_1 = \{&4,5,8,9,12,13,15,20,21,22,23,24,25,26,29,31,32,35,37,38,40,41,44,54,57,58,60,\\ &61,63,66,67,69,72,73,74,75,76,77,78,83,85,86,89,90,93,94\},\\ R_5 = \{&1,4,8,9,12,13,14,16,17,18,20,24,26,27,28,29,30,31,37,38,39,40,41,42,45,47,55,\\ &57,60,61,62,63,64,65,71,72,73,74,75,76,78,82,84,85,86,88,89,90,93,94\}. \end{aligned}$$

The colouring has 34 orbits on triples; 74 496 of the 147 440 triples are red. No five points of $\mathbb{F}_{97}$ span a monochromatic $K_5^{(3)}$. (An analogous colouring of $\mathbb{F}_{89}$, which already gives $R(5,5;3) \geq 90$, is in the data files.)

$R(4,6;3) \geq 68$. $q = 67$, $g = 2$,

$$\begin{aligned} R_1 &= \{2,4,7,9,11,13,15,16,17,20,21,29,30,34,37,43,46,48,57,58,61,66\},\\ R_2 &= \{1,4,9,11,14,20,22,27,29,35,37,39,40,43,44,47,50,51,55,61,62,65\}. \end{aligned}$$

There is no red $K_4^{(3)}$ and no blue $K_6^{(3)}$; 16 214 of the 47 905 triples are red. Since $67 \equiv 3 \pmod 4$, -1 is not a square and the group has odd order $67 \cdot 33$.

$R(5,5;4) \geq 38$. $q = 37$ and $C = \mathbb{F}_{37}^*$, i.e. the colouring of the 4-subsets of $\mathbb{F}_{37}$ is invariant under the full affine group $\mathrm{AGL}(1, 37)$, which acts sharply 2-transitively; every 4-subset is equivalent to one of the form $\{0, 1, t, u\}$, and the colouring is given by the list of the 309 red pairs $\{t, u\} \subseteq \mathbb{F}_{37} \setminus \{0, 1\}$ in `data/describe_R554_37.txt` (55 orbits; 34 299 of the 66 045 quadruples are red). No five points span a monochromatic $K_5^{(4)}$.

$R(5,6;4) \geq 98$. $q = 97$ and $C = \mathbb{F}_{97}^*$: the colouring of the 4-subsets of $\mathbb{F}_{97}$ is invariant under the full affine group $\mathrm{AGL}(1, 97)$, so it is determined by the colours of the sets $\{0, 1, t, u\}$; the 1752 red pairs $\{t, u\}$ (out of 4465) are listed in `data/describe_R564_97.txt`. There are 385 orbits on 4-sets and 1 359 552 of the 3 464 840 quadruples are red. There is no red $K_5^{(4)}$ and no blue $K_6^{(4)}$. (Colourings with the same symmetry type exist for every prime $67 \leq q \leq 97$ that we tried.)

4 Verification

Each colouring was checked in three independent ways.

1. A C program (`verify.c`, `verify4.c`) reads the list of red r-sets and searches, by depth-first clique extension over all vertices, for monochromatic $K_k^{(r)}$ of either colour; it reports none. It reports violations for perturbed inputs.

2. For the 3-uniform colourings a Python script checks translation invariance and all k-subsets containing 0; for $R(4,5;3)$ and $R(5,5;4)$ all k-subsets of the 37 points were checked by brute force (e.g. all $\binom{37}{5} = 435\,897$ five-sets).

3. A Lean 4 proof (below).

Formal statement. The Lean file `lean/Main.lean` (Lean 4.35.0, core library only, no Mathlib) represents a colouring of the 3-subsets of $\{0,\dots,N-1\}$ by a Boolean function on increasing triples (`true` = red) and defines, e.g.,

```
def HasMonoK4 (N : Nat) (col : Nat → Nat → Nat → Bool) (χ : Bool) : Prop :=
  ∃ a b c d : Nat, a < b ∧ b < c ∧ c < d ∧ d < N ∧
    col a b c = χ ∧ col a b d = χ ∧ col a c d = χ ∧ col b c d = χ
def HasMonoK5 (N : Nat) (col : Nat → Nat → Nat → Bool) (χ : Bool) : Prop :=
  ∃ a b c d e : Nat, a < b ∧ b < c ∧ c < d ∧ d < e ∧ e < N ∧
    col a b c = χ ∧ col a b d = χ ∧ col a b e = χ ∧ col a c d = χ ∧ col a c e = χ ∧
    col a d e = χ ∧ col b c d = χ ∧ col b c e = χ ∧ col b d e = χ ∧ col c d e = χ
```

and proves

```
theorem R453.main : ∃ col : Nat → Nat → Nat → Bool,
    ¬ HasMonoK4 37 col true ∧ ¬ HasMonoK5 37 col false
theorem R553.main : ∃ col : Nat → Nat → Nat → Bool,
    ¬ HasMonoK5 97 col true ∧ ¬ HasMonoK5 97 col false
```

which say exactly that $R(4,5;3) > 37$ and $R(5,5;3) > 97$. The theorems `R463.main` ($N = 67$, no red $K_4^{(3)}$, no blue $K_6^{(3)}$), `R554.main` and `R564.main` (colourings `col : Nat → Nat → Nat → Nat → Bool` of increasing quadruples, 5 resp. 15 quadruples per clique) are analogous, and `R453f.main` re-proves $R(4,5;3) > 37$ by a kernel search over all increasing tuples that does not use any symmetry. A second file, `lean/MainFinset.lean` (with Mathlib), restates the two 3-uniform diagonal-type results with `Finset`s, e.g.

```
theorem R453.main_finset : ∃ c : Finset (Fin 37) → Bool,
    (∀ S : Finset (Fin 37), S.card = 4 → ∃ t ∈ S.powersetCard 3, c t = false) ∧
    (∀ T : Finset (Fin 37), T.card = 5 → ∃ t ∈ T.powersetCard 3, c t = true)
```

Proof structure. The colouring is *defined* through differences, `col x y z := Fmask.testBit ((y-x)*q + (z-x))`, where the natural number `Fmask` encodes the colours of the triples $\{0, d_1, d_2\}$; therefore `col (x-a) (y-a) (z-a) = col x y z` for $a \le x < y < z$ (lemma `col_shift`, proved by `omega`), which is the formal version of Lemma 2.1 with $s = \min S$ (no wrap-around is needed). The kernel then evaluates, by `decide +kernel`, a Boolean function performing a pruned search for monochromatic cliques through 0: a vertex is only added if all new r-sets have the required colour. Two short lemmas (`noRed0`, `noBlue0`) turn the value `true` of this search into the statement that no monochromatic clique through 0 exists. In the 4-uniform files the colouring is `col x y z w :=`

`(row (y-x)).testBit ((z-x)*q + (w-x))`, where `row` is a binary decision tree returning masks of q^2 bits (this keeps the numbers handled by the kernel small), and the search is split into one declaration per second vertex. `#print axioms` reports only `propext` and `Quot.sound` for all five main theorems; no `native_decide` is used. Kernel times: on a Xeon Gold 6254 node, Lean 4.35.0, one core: the whole file `lean/Main.lean` needs 4595 s CPU (1 h 21 min wall) and 13.7 GB, most of it for `R564` (individually, on another machine: `R453` 5 s, `R453f` 53 s, `R553` 331 s, `R463` 128 s, `R554` 26 s); the full output, including the `#print axioms` lines, is in `lean/lean_output.txt`.

5 Searches that did not succeed

For completeness we record the parameter ranges in which no colouring was found ($m = |C|$) ("UNSAT" means the SAT solver proved that no colouring with the stated symmetry exists).

- $R(5,5;3)$, C = squares: SAT for $q = 89, 97$; UNSAT for $q = 101, 103, 107, 109, 113, 121, 125$. Smaller C ($m \in \{25, 34\}$) at $q = 101, 103$ and the one-vertex extension of the 97-point colouring (a SAT problem with 4656 variables) were not decided within the time limits.
- $R(4,6;3)$, C = squares: SAT for $q = 67$; UNSAT for $q = 71, 73, 79, 81, 83, 89, 97$. Also UNSAT for $q = 71$ with $m \in \{10, 14\}$ and $q = 73$ with $m \in \{8, 12\}$.
- $R(5,5;4)$: SAT for $q = 37$ ($m = 36, 18$); UNSAT for $q = 41$ ($m = 40, 20$). For $R(5,6;4)$ every prime $67 \le q \le 103$ tried was SAT.
- $R(4,5;3)$: SAT for $q = 37$, $m = 3$; UNSAT for $q = 37$ with $m \in \{2, 4, 6, 9, 12, 18, 36\}$, for $q = 41$ with $m \in \{2, 4, 5, 8, 10, 20, 40\}$, for $q \in \{43, 47, 49, 53\}$ with large m, and for dihedral colourings with $34 \le n \le 37$. Not decided within the time limits: $q = 43, 49, 61, 67$ with $m = 3$, the cyclic group on 38 points, $\mathbb{Z}_{19} \rtimes C_3$ acting on two copies of $\mathbb{F}_{19}$ (38 points), and the one-vertex extension of the 37-point colouring (666 variables). The extension of the 37-point family by a fixed point ($G(37, C_3)$ acting on 38 points) is UNSAT, as are its analogues with 2 and 3 fixed points and with $|C| \in \{6, 9\}$.
- $R(4,4,4;3)$ (three colours): UNSAT for C of index 2 and 3 for $89 \le q \le 125$.

These negative results only concern the very symmetric families searched; they say nothing about $R(k,l;r)$ itself.

6 Remarks and limitations

- The theorems are lower bounds only; no upper bounds are claimed. All verification is of explicit finite objects; novelty is claimed to the best of our knowledge as of October 14, 2026.
- By a general product-type construction quoted in [1, 7.1(a)] (attributed there to [5]), a 2-colouring witnessing $R(5,5;3) > m$ yields $R_k(5;3) > m^{2^{k-2}}$; with $m = 97$ this would give $R(5,5,5;3) \ge 9410$. We did not formalise this consequence.
- The sweep is far from exhaustive: smaller subgroups C, non-prime orders, and groups with fixed points were explored only partially, and several instances were stopped by a time limit; larger bounds are plausible. For $R(5,6;4)$ every prime $67 \le q \le 103$ that we tried admits an $\mathrm{AGL}(1,q)$-invariant colouring; the colourings for $q = 101$ and $q = 103$ (giving

$R(5,6;4) \geq 104$) were verified by the C program only, not in Lean, because the kernel check grows like q^5.

- We also tried, without success, to decide $R(K_4 - t, K_5; 3) \in \{14, 15, 16\}$ by SAT: local search never gets below six violated constraints on 14 points (while 13 points are easy), which suggests $R(K_4 - t, K_5; 3) = 14$, but we have no proof.

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